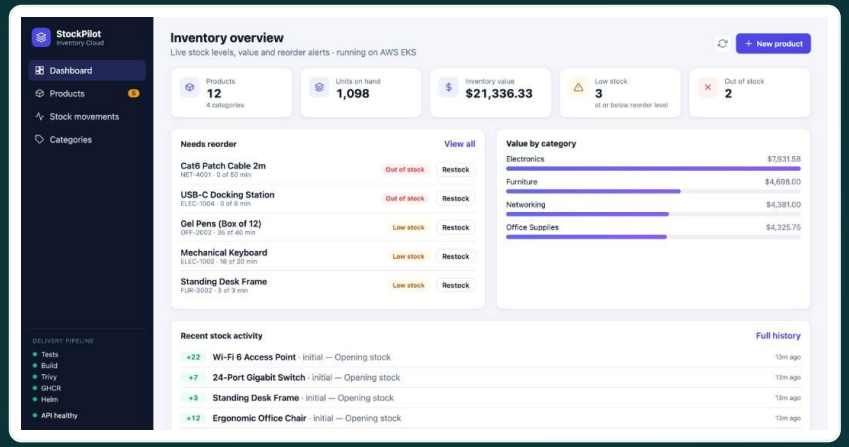
StockPilot

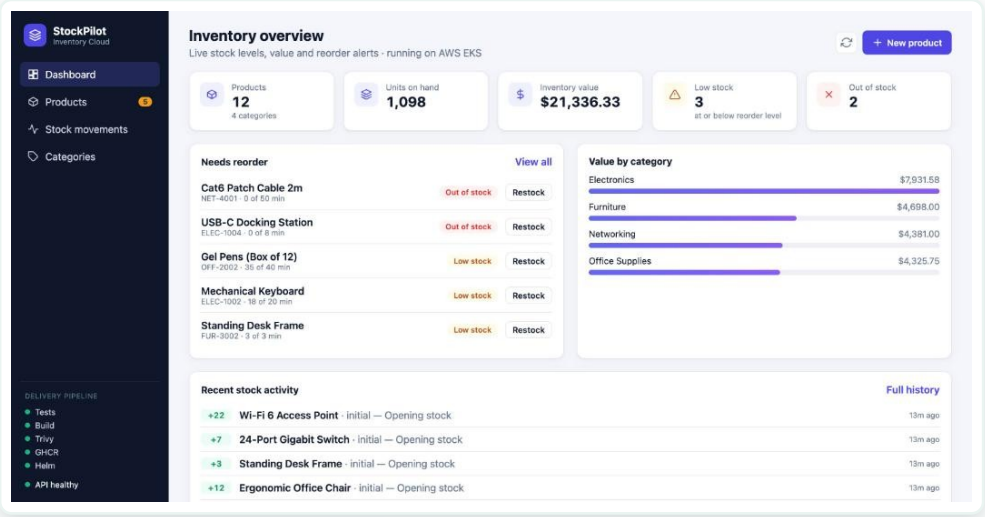
Inventory management, shipped from a Git push to a monitored Kubernetes cluster on AWS



Pragya Tripathi · Roll No 10032

DevOps Final Capstone · October 2026

The application



Live on AWS EKS, reached through the Ingress load balancer

React + Vite frontend

Dashboard, products, stock movements, categories; works on phones

FastAPI backend

14 REST endpoints plus /health, /ready and /metrics

PostgreSQL + Alembic

3 tables, versioned migrations, demo seed data

21 automated tests

pytest on in-memory SQLite, plus ruff lint, on every push

From commit to cluster



GitHub Actions

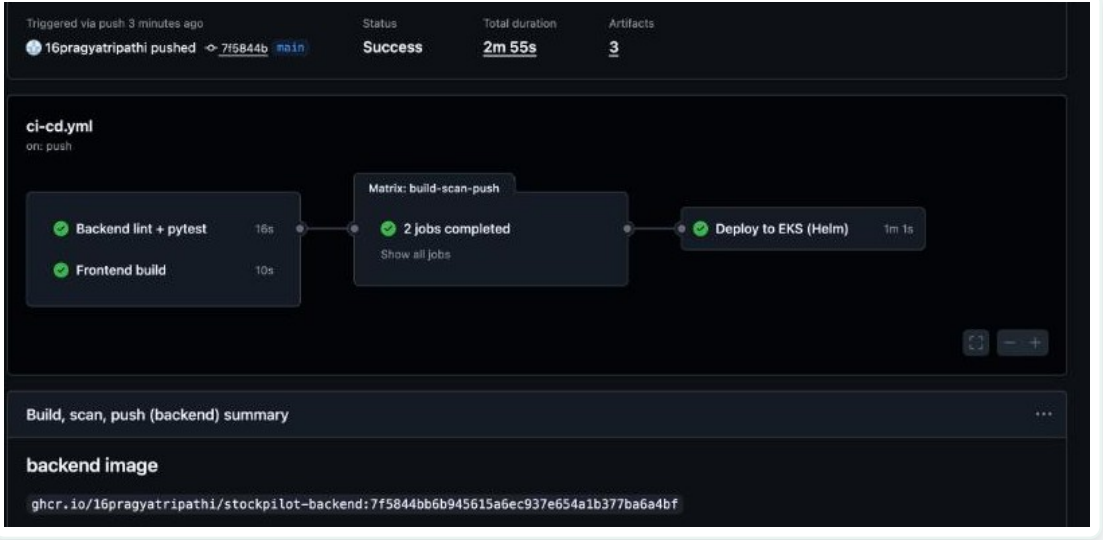
AWS, built by Terraform

Commit SHA
is the image tag, so every running pod traces back to one commit

Quality gate
a failed test or a fixable HIGH/CRITICAL CVE stops the release

No stored keys
GitHub OIDC assumes a short-lived AWS role to deploy

CI/CD and security



A real run: every job green, ending in the Helm deploy to EKS

5 / 5 jobs
green, ending in the EKS deploy

0
fixable HIGH/CRITICAL CVEs in both images

Non-root
backend uid 10001, frontend uid 101

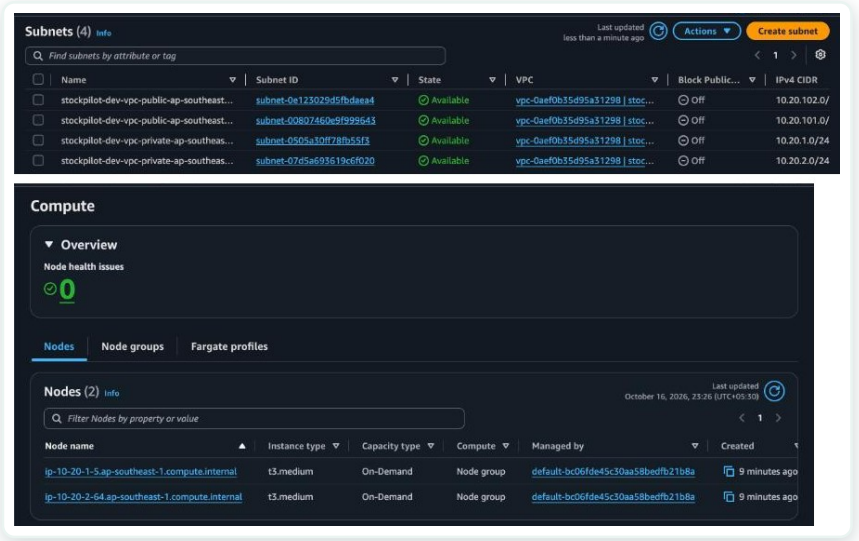
Infrastructure as code

VPC across 2 AZs
Public subnets + NAT gateway, private subnets for nodes

EKS 1.36 + managed nodes
2 × t3.medium; VPC CNI, CoreDNS, EBS CSI via Pod Identity

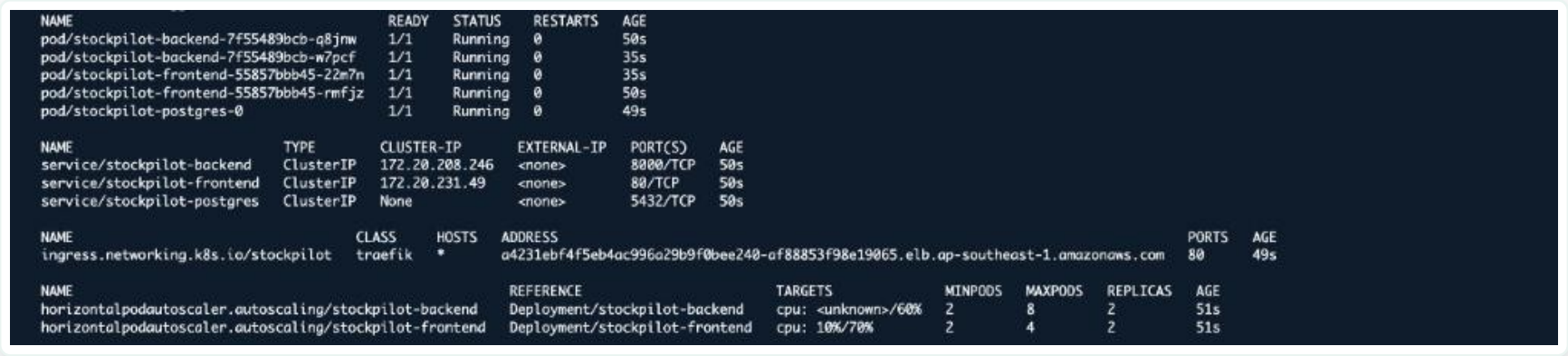
GitHub OIDC deploy role
Only the main branch of this repo can assume it

64 resources, one plan
terraform plan, apply, and destroy after the demo



AWS console: the Terraform VPC's 4 subnets and the 2 EKS nodes

Kubernetes and Helm on EKS



```
NAME                                        READY   STATUS    RESTARTS   AGE
pod/stockpilot-backend-7f55489bcb-q8jnw     1/1     Running   0          50s
pod/stockpilot-backend-7f55489bcb-w7pcf     1/1     Running   0          35s
pod/stockpilot-frontend-55857bbb45-22m7n    1/1     Running   0          35s
pod/stockpilot-frontend-55857bbb45-rmfjz    1/1     Running   0          50s
pod/stockpilot-postgres-0                   1/1     Running   0          49s

NAME                           TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
service/stockpilot-backend     ClusterIP   172.20.208.246   <none>        8000/TCP   50s
service/stockpilot-frontend    ClusterIP   172.20.231.49    <none>        80/TCP     50s
service/stockpilot-postgres    ClusterIP   None             <none>        5432/TCP   50s

NAME                                    CLASS     HOSTS   ADDRESS                                                                                  PORTS   AGE
ingress.networking.k8s.io/stockpilot    traefik   *       a4231ebf4f5eb4ac996a29b9f0bee240-af88853f98e19065.elb.ap-southeast-1.amazonaws.com   80      49s

NAME                                                                REFERENCE                          TARGETS              MINPODS   MAXPODS   REPLICAS   AGE
horizontalpodautoscaler.autoscaling/stockpilot-backend    Deployment/stockpilot-backend    cpu: <unknown>/60%   2         8         2          51s
horizontalpodautoscaler.autoscaling/stockpilot-frontend   Deployment/stockpilot-frontend   cpu: 10%/70%         2         4         2          51s
```

Helm chart

2+ backend and frontend pods, PostgreSQL StatefulSet on a gp3 volume

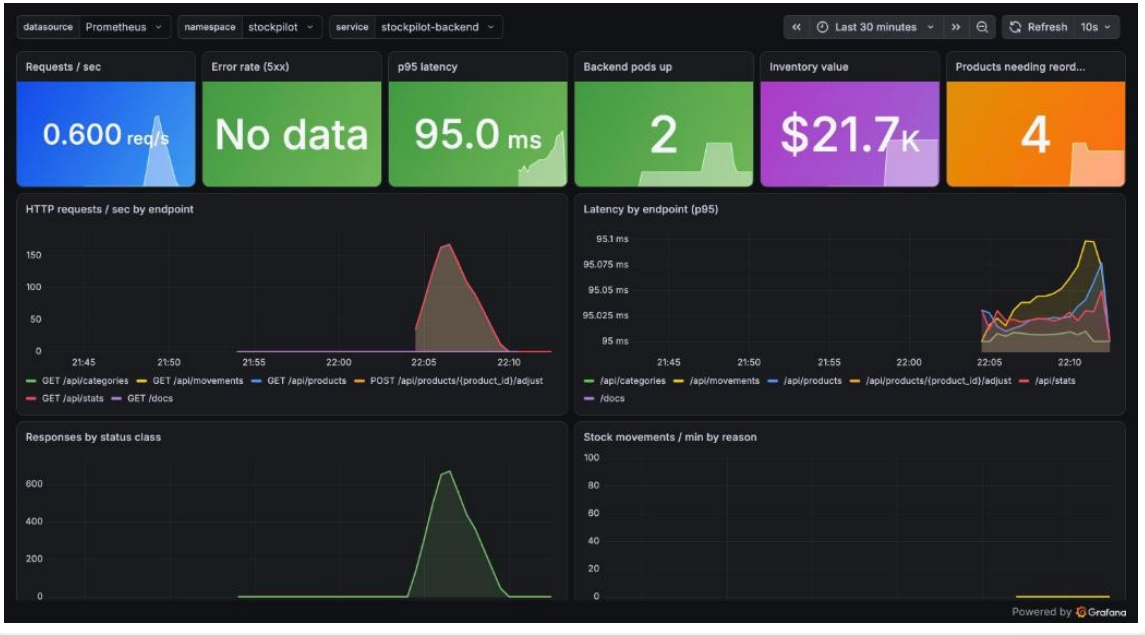
Ingress

/ to React, /api to FastAPI, through an AWS Network Load Balancer

Self-healing

startup, readiness and liveness probes, plus PodDisruptionBudgets

Autoscaling and monitoring



The Grafana dashboard ships with the Helm chart and loads automatically

2 → 6 pods
HPA scaled the backend under load

~660 req/s
peak traffic during the load test

~95 ms
p95 latency, scraped by Prometheus

Troubleshooting lab

```
$ kubectl get pods -l lab=broken-image
broken-image-695b…   0/1   ImagePullBackOff
$ kubectl describe pod -l lab=broken-image
Failed to pull …:v9.9.9-does-not-exist: not found
$ kubectl set image deploy/… backend=<real tag>
broken-image-585f…   1/1   Running

$ kubectl get endpointslices … api-broken
stockpilot-api-broken   <unset>   (no endpoints)
$ kubectl patch svc … selector=stockpilot-backend
stockpilot-api-broken   8000   10.244.1.23,10.244.1.14
```

ImagePullBackOff

Wrong tag; describe pod names it; set image fixes it

Service with no endpoints

Selector didn't match the pod labels; patch the selector

Running but 0/1 Ready

/ready returns 503 on a bad DB password; no traffic sent

Every step automated and observable

21
automated tests

0
HIGH/CRITICAL CVEs

64
AWS resources

2 → 6
pods autoscaled

github.com/16pragyatripathi/devops-capstone-stockpilot

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